

TRUNG TUAN MAI

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EDUCATION

Bachelor of Science, Software Engineering

Expected April 2027

Schulich School of Engineering, University of Calgary

CORE SKILLS

Programming: Python, Java

Data Analysis: pandas, NumPy, data cleaning, feature engineering

Machine Learning: scikit-learn, regression modeling, model evaluation

Visualization: matplotlib, seaborn

Databases: MongoDB, MySQL

Tools: Git, Docker, Linux

Concepts: statistical analysis, data preprocessing, model evaluation, data pipelines

PROJECTS

Predicting Car Sales Revenue – Machine Learning System

(Machine Learning Course)

December 2025

- Engineered a full machine learning workflow in Python using pandas, NumPy, and scikit-learn to process a dataset exceeding **500K records**.
- Performed data cleaning, missing-value handling, and categorical encoding to prepare structured datasets for modeling.
- Engineered 15+ predictive features using temporal extraction and statistical correlation analysis to improve predictive performance.
- Trained and evaluated multiple regression models (Linear Regression, Decision Tree, Random Forest, Gradient Boosting) using RMSE, MAE, and R^2 .
- Achieved $R^2 = 0.91$ and selected optimal models through comparative analysis across 47+ controlled experiments.

Data Ingestion & Backend System

(Web-Based Systems Course)

February 2026

- Designed a data ingestion pipeline to collect and normalize multi-source restaurant datasets into a structured MongoDB database.
- Implemented data validation and deduplication logic to ensure consistency and reliability of analytical datasets.
- Developed workflows for extracting, transforming, and storing structured data to support downstream analysis.

Backend Job-Market Simulation System

(Software Architecture Course)

February 2026

- Modeled structured datasets representing user progression, application history, and hiring outcomes in a relational database.
- Designed rule-based logic to simulate hiring probability based on resume score and progression factors.
- Generated structured outputs that enable analysis of decision-making behavior across simulated job-market scenarios.

EXTRA-CURRICULARS

Financial Transaction Processing Backend

JP Morgan Chase Virtual Experience in Forage

January 2026

- Built backend data services using Spring Boot to manage account data persistence and retrieval.
- Processed asynchronous transaction data streams using Kafka to simulate event-driven data flows.

WORK EXPERIENCE

Server - Pho Pham Restaurant

Present

- Managed high-volume transactional data (100+ per shift), maintaining accuracy and consistency in fast-paced operations.
- Applied structured problem-solving to resolve real-time operational issues and maintain workflow efficiency.